

Simulation Report

DWSIM v.9.0

Details

Title:MySimulation_45

Comments:

Object:Feed

Type:Material Stream

Property	Value	
Temperature	298.15	K
Pressure	101325	Pa
Mass Flow	1	kg/s
Molar Flow	11.7474	mol/s
Volumetric Flow	0.00115109	m3/s
Mixture Molar Fraction		
Benzene	0.5	
Toluene	0.5	
Density (Mixture)	868.745	kg/m3
Molecular Weight (Mixture)	85.1251	kg/kmol
Specific Enthalpy (Mixture)	-409.452	kJ/kg
Specific Entropy (Mixture)	-1.04955	kJ/[kg.K]
Molar Enthalpy (Mixture)	-34854.6	kJ/kmol
Molar Entropy (Mixture)	-89.3434	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.137856	W/[m.K]
Molar Fraction (Vapor)		
Benzene	0.770659	
Toluene	0.229341	
Molar Fraction (Overall Liquid)		
Benzene	0.5	
Toluene	0.5	
Molar Fraction (Liquid 1)		
Benzene	0.5	
Toluene	0.5	
Molar Fraction (Liquid 2)		
Benzene	0	
Toluene	0	
Molar Fraction (Aqueous)		
Benzene	0	
Toluene	0	
Molar Fraction (Solid)		
Benzene	0	
Toluene	0	

Object:Distillate

Simulation Report

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Details

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Comments:

Temperature	362.232	K
Pressure	101325	Pa
Mass Flow	0.815727	kg/s
Molar Flow	9.74741	mol/s
Volumetric Flow	0.00101451	m3/s
Mixture Molar Fraction		
Benzene	0.60256	
Toluene	0.39744	
Density (Mixture)	804.061	kg/m3
Molecular Weight (Mixture)	83.6866	kg/kmol
Specific Enthalpy (Mixture)	-302.888	kJ/kg
Specific Entropy (Mixture)	-0.737849	kJ/[kg.K]
Molar Enthalpy (Mixture)	-25347.6	kJ/kmol
Molar Entropy (Mixture)	-61.748	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.120511	W/[m.K]
Molar Fraction (Vapor)		
Benzene	0	
Toluene	0	
Molar Fraction (Overall Liquid)		
Benzene	0.60256	
Toluene	0.39744	
Molar Fraction (Liquid 1)		
Benzene	0.60256	
Toluene	0.39744	
Molar Fraction (Liquid 2)		
Benzene	0	
Toluene	0	
Molar Fraction (Aqueous)		
Benzene	0	
Toluene	0	
Molar Fraction (Solid)		
Benzene	0	
Toluene	0	

Object: Bottoms

Type: Material Stream

Simulation Report

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Comments:

Temperature	383.814	K
Pressure	101325	Pa
Mass Flow	0.184273	kg/s
Molar Flow	2	mol/s
Volumetric Flow	0.000236356	m3/s
Mixture Molar Fraction		
Benzene	0.000153466	
Toluene	0.999847	
Density (Mixture)	779.641	kg/m3
Molecular Weight (Mixture)	92.1363	kg/kmol
Specific Enthalpy (Mixture)	-253.012	kJ/kg
Specific Entropy (Mixture)	-0.619582	kJ/[kg.K]
Molar Enthalpy (Mixture)	-23311.5	kJ/kmol
Molar Entropy (Mixture)	-57.086	kJ/[kmol.K]
Thermal Conductivity (Mixture)	0.111754	W/[m.K]
Molar Fraction (Vapor)		
Benzene	0	
Toluene	0	
Molar Fraction (Overall Liquid)		
Benzene	0.000153466	
Toluene	0.999847	
Molar Fraction (Liquid 1)		
Benzene	0.000153466	
Toluene	0.999847	
Molar Fraction (Liquid 2)		
Benzene	0	
Toluene	0	
Molar Fraction (Aqueous)		
Benzene	0	
Toluene	0	
Molar Fraction (Solid)		
Benzene	0	
Toluene	0	

Object: DCOL-1

Type: Distillation Column

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Comments:

Condenser Pressure Drop	0	Pa
Condenser Duty	1103.98	kW
Reboiler Duty	-1219.73	kW
Number of Stages	20	
Column Pressure Drop	0	Pa
Stage_Pressure_1	101325	Pa
Stage_Pressure_2	101325	Pa
Stage_Pressure_3	101325	Pa
Stage_Pressure_4	101325	Pa
Stage_Pressure_5	101325	Pa
Stage_Pressure_6	101325	Pa
Stage_Pressure_7	101325	Pa
Stage_Pressure_8	101325	Pa
Stage_Pressure_9	101325	Pa
Stage_Pressure_10	101325	Pa
Stage_Pressure_11	101325	Pa
Stage_Pressure_12	101325	Pa
Stage_Pressure_13	101325	Pa
Stage_Pressure_14	101325	Pa
Stage_Pressure_15	101325	Pa
Stage_Pressure_16	101325	Pa
Stage_Pressure_17	101325	Pa
Stage_Pressure_18	101325	Pa
Stage_Pressure_19	101325	Pa
Stage_Pressure_20	101325	Pa
Stage_Efficiency_1	1	
Stage_Efficiency_2	1	
Stage_Efficiency_3	1	
Stage_Efficiency_4	1	
Stage_Efficiency_5	1	
Stage_Efficiency_6	1	
Stage_Efficiency_7	1	
Stage_Efficiency_8	1	
Stage_Efficiency_9	1	

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Comments:

Stage_Efficiency_10	1	
Stage_Efficiency_11	1	
Stage_Efficiency_12	1	
Stage_Efficiency_13	1	
Stage_Efficiency_14	1	
Stage_Efficiency_15	1	
Stage_Efficiency_16	1	
Stage_Efficiency_17	1	
Stage_Efficiency_18	1	
Stage_Efficiency_19	1	
Stage_Efficiency_20	1	
Stage_Temperature_1	362.232	K
Stage_Temperature_2	368.659	K
Stage_Temperature_3	373.159	K
Stage_Temperature_4	375.607	K
Stage_Temperature_5	376.766	K
Stage_Temperature_6	377.279	K
Stage_Temperature_7	377.5	K
Stage_Temperature_8	377.594	K
Stage_Temperature_9	377.633	K
Stage_Temperature_10	377.65	K
Stage_Temperature_11	377.657	K
Stage_Temperature_12	380.631	K
Stage_Temperature_13	382.259	K
Stage_Temperature_14	383.078	K
Stage_Temperature_15	383.473	K
Stage_Temperature_16	383.659	K
Stage_Temperature_17	383.746	K
Stage_Temperature_18	383.787	K
Stage_Temperature_19	383.806	K
Stage_Temperature_20	383.814	K
Condenser_Specification_Value	2.5	
Reboiler_Specification_Value	2	
Global_Stage_Efficiency	G6	
Condenser_Calculated_Value	2.5	
Reboiler_Calculated_Value	2	
Stream 'Feed' Stage Index	10	
Stream 'Distillate' Stage Index	0	

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Stream 'Bottoms' Stage Index	19	
Estimated Height	11000	mm
Estimated Diameter	1400.01	mm

Object: E1

Type: Energy Stream

Property	Value	
Energy Flow	1103.98	kW

Object: E2

Type: Energy Stream

Property	Value	
Energy Flow	1219.73	kW